

YAKUBU ADAM

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CAREER OBJECTIVE

Motivated toxicologist, lecturer and drug discovery scientist with research interest in neuropharmacology, medicinal chemistry, translational drug development and environmental toxicology. My goal is to integrate experimental and computational approaches to advance therapeutics for neurotoxicity, addiction, and neurodegenerative disorders

RESEARCH INTERESTS

- Neuropharmacology and Neurotoxicity
- Medicinal and combinatorial chemistry in Drug discovery
- Small-Molecule Drug Discovery for CNS
- Metal-Induced Neurotoxicity and Aging-Related Vulnerability
- Mechanistic Toxicology of Environmental Exposures and Neurodegeneration

EDUCATIONAL BACKGROUND

- Klintaps University College of Health and Allied Sciences, Ghana **Pending Certification**
Master of Public Health (MPH)
Specialisation: Epidemiology and Disease Control
Thesis: Assessing the perspective of health care workers on barriers affecting childhood immunization: A case study of kokrobite and bortianor health centers, Ghana
- Ghana Institute of Management and Public Administration (GIMPA), Ghana **Pending Certification**
MSc Occupational Safety, Health and Environmental Management
Focus: Occupational Toxicology
Thesis: Assessing the role of awareness of campus safety policies and regulations on safety behavior and compliance among tertiary students in Ghana: A case study of Klintaps University
- University of Cape Coast, Ghana **Awarded March 2022**
Master of Philosophy; Drug Discovery and Toxicology
Thesis: Integrated In-Vitro and In-Silico Investigation of α -Amylase and α -Glucosidase Inhibition by Flavonoids from *Indigofera arrecta* Leaf Extracts.
- University of Cape Coast, Ghana **Awarded May 2018**
BSc Molecular biology and Biotechnology
Thesis: In-Vitro Evaluation of the Anticancer Effects of *Solanum melongena* Leaf Extracts on MCF-7 Breast Cancer Cell Lines and Phytochemical Characterization.

RESEARCH (NON-TEACHING) EXPERIENCE

Toxicologist **February 2021 - Present**

Forensic Investigation for National Development-Ghana (FIND-GH), Accra, Ghana

- Implemented and monitored workplace drug testing programs, ensuring compliance with occupational health standards, ethical guidelines, and regulatory requirements.
- Evaluated herbal and natural products for toxic adulterants, including xeno-aphrodisiac substances, using toxicological screening and analytical methods.
- Assessed health risks associated with adulterated products and provided evidence-based recommendations to support consumer safety and regulatory enforcement.

Graduate Research Assistant

August 2019 - 2022

University of Cape Coast, School of Allied Health Sciences, Department of Biomedical Sciences (Medicinal biochemistry), Accra, Ghana

Supervisor: Professor Michael Adinortey

- Performed extraction and characterization of bioactive compounds from *Indigofera arrecta*, *Strychnos spinosa* and *Solanum melongena* and many others for phytochemical and pharmacological analysis.
- Analytical and natural product chemistry, encompassing phytochemical analysis and instrumental techniques such as HPLC, GC-MS, NMR, and mass spectrometry.
- Pharmacological and toxicological research, including in vivo animal studies for efficacy, dose–response, and safety evaluation.
- Maintained detailed laboratory notebooks, analyzed data using statistical software, and prepared reports and presentations summarizing findings.
- Maintained a safe working environment by handling hazardous chemicals appropriately and ensuring compliance with institutional and governmental regulations.
- Provided guidance to undergraduate students and new lab members on experimental techniques and data analysis.

Research Assistant**September 2018 - 2019**

University of Cape Coast, School of Biological Sciences, Department of Molecular Biology and Biotechnology

Supervisor: Professor Foster Kyei

- Conducted in vitro anticancer studies by evaluating the cytotoxic effects of *Solanum melongena* leaf extracts on MCF breast cancer cell lines.
- Performed phytochemical screening and characterization of plant extracts to identify bioactive secondary metabolites.
- Analyzed and interpreted experimental data to establish relationships between phytochemical composition and observed anticancer activity.

Research Intern**May – August 2017**Council for Scientific and Industrial Research (CSIR), Ghana. Animal Research Institute, Accra, Ghana
Department of Microbiology

- Designed and conducted microbiological experiments, including culture, isolation, identification, and characterization of microorganisms.
- Performed antimicrobial susceptibility and drug–microbe interaction studies to evaluate the efficacy and resistance patterns of therapeutic agents.
- Applied molecular microbiology techniques (e.g., PCR, electrophoresis, sequencing) for microbial detection, typing, and genetic analysis.
- Analyzed, documented, and disseminated research findings through data analysis, report writing, and scientific presentations or publications.

Research Intern**May-August 2016**

Lapaz Community Hospital, Accra, Ghana

- Collected and analyzed clinical body fluid samples to detect abnormal components using standard laboratory diagnostic techniques.
- Interpreted test results, ensured data accuracy, and prepared clear documentation and laboratory reports.
- Maintained electronic laboratory records and assisted senior laboratory scientists during clinical and analytical procedures.

TEACHING EXPERIENCE

Adjunct Lecturer

Jan 2026 - Present

Pentecost University, Faculty of Health Sciences, Department of Herbal Medicine, Accra, Ghana

- Taught undergraduate courses in Forensic medicine and toxicology, pharmacology, and safety evaluation of herbal medicines, fostering critical thinking and practical skills.
- Supervised student research projects and laboratory practical, guiding experimental design, data analysis, and reporting in toxicological studies.
- Conducted research on herbal product safety, evaluating bioactive compounds, potential toxicants, and their pharmacological effects.
- Developed and updated course curricula and training materials, integrating current advances in toxicology, herbal medicine, and regulatory standards.

Lecturer (Full-time)

Jan 2025 - Present

Klintaps University of Health and Allied Sciences, Department of Medical Laboratory Sciences, Accra.

- Taught courses in clinical toxicology, pharmacology, and drug metabolism, equipping students with practical and theoretical knowledge.
- Supervised laboratory sessions and student research projects, ensuring proper experimental design, data collection, and analysis.
- Conducted research on chemical, microbial, and herbal toxicants, evaluating their effects on human health and laboratory safety.
- Developed and maintained laboratory protocols and safety standards, promoting compliance with Good Laboratory Practice (GLP) and occupational health guidelines.

Adjunct Lecturer

Oct 2024 – Present

Islamic Nursing Training School, Department of General Nursing, Accra, Ghana

- Delivered lectures and tutorials in Pharmacology, Therapeutics, Pharmacovigilance, and Basic Statistics, fostering student understanding and application of core nursing concepts.
- Supervised and mentored students in practical sessions, research projects, and clinical case studies, ensuring competency in drug administration, patient safety, and data analysis.
- Assessed and evaluated student performance, including assignments, quizzes, and examinations, providing constructive feedback to support academic growth.
- Developed and updated course materials and teaching aids, integrating current evidence-based practices, drug safety guidelines, and statistical methodologies.

Adjunct Lecturer

Sept 2024 – Oct. 2025

Maranatha University College, Department of Pharmaceutical Sciences, Accra Ghana

- Taught core courses including Therapeutics, Forensic Pharmacy, Pharmacognosy, and Pharmaceutical Chemistry, ensuring students gain practical and theoretical expertise.
- Supervised student laboratory work and research projects, promoting skills in drug formulation, analysis, and quality control.
- Assessed and evaluated student performance, including assignments, quizzes, and examinations, providing constructive feedback to support academic growth.

Adjunct Lecturer

Sept 2023 – Jan. 2025

Pharmatrust Professional College, Department of Pharmaceutical Sciences, Accra Ghana

- Supervised student laboratory work and research projects, promoting skills in drug formulation, analysis, and quality control.

- Assessed and evaluated student performance, including assignments, quizzes, and examinations, providing constructive feedback to support academic growth
- Taught core courses including Therapeutics, Forensic Pharmacy, Pharmacognosy, and Pharmaceutical Chemistry, ensuring students gain practical and theoretical expertise.

Senior Research Tutor

Oct 2023 – Oct. 2024

University of Cape Coast-Switch Program, College of Distance Education, Accra, Ghana

- Delivered lectures and practical sessions in Cytology, Genetics, Anatomy, Physiology, and Cell & Tissue Organization, fostering foundational understanding for undergraduate students.
- Supervised and guided laboratory exercises and tutorials, ensuring students develop hands-on skills and proper experimental techniques.
- Assessed student learning outcomes, including grading assignments, quizzes, and practical exams, providing constructive feedback to enhance academic performance.
- Developed and updated course content and teaching resources, integrating current scientific literature and best practices in biomedical education.

Teaching Assistant (National Service)

Sept. 2018 – August. 2019

University of Cape Coast, Department of Molecular Biology and Biotechnology, Ghana

- Delivered lectures and practical sessions in Developmental biology, Molecular Genetics and Biodiversity of living things
- Supervised and guided laboratory exercises and tutorials, ensuring students develop hands-on skills and proper experimental techniques.
- Assessed student learning outcomes, including grading assignments, quizzes, and practical exams, providing constructive feedback to enhance academic performance.

LABORATORY AND COMPUTATIONAL SKILLS

Laboratory Skills

- Analytical chemistry techniques (NMR, GC-MS, HPLC, Mass Spectrometry)
- Natural product chemistry and Phytochemistry
- Toxicological and pharmacological studies using animal models
- Pharmacovigilance (Signal generation)
- Quality Control (HACCP, GMP, GLP and GCP)
- Molecular biology (PCR, western blotting, DNA Extraction and sequencing, etc.)
- Microbiological techniques (Drug-microbial testing, cell culture, etc)

Computational Skills

- Data Analytics, Visual analytics and Machine Learning
- Computational drug discovery (Spartan, Chemdraw, PYRX, Discovery studio and Autodock tool)
- Bioinformatics (R studio, Benchling, MEGA and NCBI)
- Scientific research data analysis (SPSS, Stata, Tableau & GraphPad Prism)
- Windows and Linux operating systems for Molecular Dynamic (MD) simulations

PROFESSIONAL AFFILIATIONS

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| • Ghana Science Association – Member | 2025 - Present |
| • National Association of Safety and Environmental Professionals – Member | 2026 - Present |

LECTURES, WORKSHOPS, AND PROJECTS

WORKSHOPS

- 2024 **Title:** AI/2050 AI/ML in Drug Discovery Workshop
H3D Foundation and Ersilia
Role: Participant
Venue: University of Ghana
- 2023 **Title:** Structural Elucidation of Natural product (NMR)
Department of Chemistry
University of Ghana (UG)
Role: Participants
Venue: WACCBIP (UG)
- 2022 **Title:** Bioinformatics for Biomedical Research UCCSMS Bioinformatics Workshop
Department of Medical biochemistry,
University of Cape Coast.
Role: Participant
Venue: UCCSMS Clinical Teaching Centre
- 2022 **Title:** Molecular Docking
Department of Chemistry, University of Cape Coast (UCC).
Role: Participant
Venue: Graduate Research Commons-Main library, UCC

LECTURES

- **Topic:** Ghana's Hidden Threat: Lead Poisoning's Silent Impact
Role: Speaker
Venue: Airport View Hotel, Accra-Ghana (2026)
- **Topic:** Kidney failure, and the implications of illegal mining (Galamsey) activities in Ghana (Organized by University for Development-UDS Bioscience Lecture series)
Role: Speaker
Venue: Nyankpala Campus
- **Topic:** Illegal mining (Galamsey) and the sudden rise in kidney failure (Organized by Forensic Investigation for National Development, Ghana)
Role: Speaker

Manuscripts Under review

- Computational prioritization of FSP1 inhibitor candidates targeting the ferroptosis-associated protein interactome (In-silico Research in Biomedicine)
- Toxic Exposures and Reproductive Health in Ghana: Unmasking the Environmental Threat to Fertility, Pregnancy, and Fetal Development (Birth Defects Research Journal)
- The Use of Artificial Intelligence in Advancing Molecular Biology in Africa (Molecular Genetics and Genomics).
- Hepatoprotective Effects of Vernonia amygdalina Leaf Extract: In Vivo and In silico Evidence Against Ethanol-Induced Liver Injury (In Silico Pharmacology)

VOLUNTEERING AND MEDIA ENGAGEMENT

TV3 Ghana (National Television)

Topic: Lead Poisoning in Ghana

<https://www.facebook.com/share/v/1GY9LD3i24/>

TV3 Ghana (National Television)

Program: NewDay show

Topic: Illegal mining and its health implications

<https://www.facebook.com/share/v/pR993yLJAjTYcNsK/?mibextid=oFDknk>

Ghana Television (State Owned Television)

Program: Breakfast Show

Topic: Illegal mining and its associated congenital anomalies

<https://www.facebook.com/share/v/6psPrvRLnaATXgmU/?mibextid=oFDknk>

United Television (National Television)

Program: Healthline (Adekye Nsroma)

Topic: Galamsey (illegal mining), a risk factor of kidney disease?

<https://www.youtube.com/watch?v=uQ9faJQ1Qwc&pp=ygUYYYWRla3llIG5zcm9tYSAxNi8wNC8yMDI0>

Health and Safety Television (HSTV)

Program: Stay Alive

Topic: Galamsey (illegal mining) and the surge in kidney failure

<https://www.facebook.com/hstvlivegh/videos/3269765929995330>

Peace FM (Widely listened Radio Station)

Program: Akan news at 12 (Midday news)

Topic: Mercury usage and nephrotoxicity

https://www.youtube.com/watch?v=tWaj_-FYoqQ

Neat FM (Widely listened radio station)

Program: MeMAN Nti

Topic: Galamsey (illegal mining) and the incidence of kidney failure in Ghana

<https://www.facebook.com/NEAT100.9FM/videos/322193770639141/>

REFERENCES

- Prof. Aaron T. Asare || Professor – Plant Biotechnology || University of Cape Coast, Cape Coast, Ghana || aasare@ucc.edu.gh || +233244954045 || Relationship: Supervisor.
- Prof. Micahel Adinortey || Professor of Medicinal Biochemistry || University of Cape Coast, Cape Coast, Ghana || madinortey@ucc.edu.gh || +233506373607 || Relationship: Supervisor.
- Prof. Martins Ekor || Professor of Toxicology || University of Cape Coast, Cape Coast, Ghana || martins.ekor@ucc.edu.gh || +233247950762 || Relationship: Academic Lecturer.